

SKANDESH GOUTHAM RANGANATHAN J

Software Development Engineer

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[LinkedIn](#) | [LeetCode](#) | [GitHub](#)

EDUCATION

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science (CGPA: 8.29)

Aug 2023 – May 2027

Chennai, India

TECHNICAL SKILLS

- **Languages:** Python, C, Java, JavaScript, HTML/CSS, SQL
- **Frameworks:** FastAPI, Node.js, Express.js, React.js, TensorFlow, Flask
- **Tools:** Git, AWS, PostgreSQL, MySQL, Docker, **Kali Linux**, Postman, Cisco Packet Tracer
- **Concepts:** Data Structures, RSA & SHA-256, JWT, REST APIs, **Computer Networks**, Cloud Computing

KEY PROJECTS

SecureChain: Digital Evidence Custody

FastAPI, PostgreSQL, SHA-256

[GitHub](#)

- Architected a cybersecurity platform using Dual-Chain Hashing (DCH-IV) to ensure 100% integrity of digital evidence without blockchain overhead.
- Engineered a cross-chain binding mechanism with RBAC and JWT authentication, reducing forensic verification latency by 40% across RESTful nodes.

Vault: AI Financial Management

Flask, Python, JavaScript

[GitHub](#)

- Developed a full-stack financial application featuring automated transaction tracking and a regression-based engine to forecast spending, improving prediction accuracy by **[91]%**.
- Integrated a conversational AI assistant (Penny) with gamified alerts, increasing user engagement by **[90]%** and reducing budget overruns by **[45]%** through data-driven alerts.

OfferCred: Certificate Verification

Node.js, Express, PostgreSQL

[GitHub](#)

- Built an RSA-signature platform that authenticated 200+ certificates and successfully blocked 50+ fraud attempts using custom validation logic.
- Implemented Bcrypt and JWT-based security protocols, resulting in a 99% reduction in unauthorized system access incidents during testing phases.

Cloud Load Forecaster (HDRL-LF)

Python, TensorFlow, LSTM

[GitHub](#)

- Engineered a hybrid VM placement framework processing 10k+ data points, achieving 92% forecasting accuracy to optimize cloud resource allocation.
- Enhanced system energy efficiency by 15% through the implementation of Dueling Deep Q-Network (DQN) agents for dynamic load balancing.

PUBLICATIONS & RESEARCH

NeuroBridge - AI Mental Health Crisis Detection

- Proposed a federated AI system for real-time crisis detection using multi-device behavioral drift analysis.

Autonomous Vehicle Security

- Analyzed vulnerabilities in autonomous systems and proposed software-based defenses against ethical hacking.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **AWS Academy Graduate:** Gained hands-on experience in AWS Cloud architecture and storage via AWS Student Platform labs.
- **Networking Basics (Cisco):** Specialized in TCP/IP protocols and connectivity troubleshooting using Cisco Packet Tracer.
- **Cloud Computing (NPTEL):** Trained in Virtualization, Resource Management, and SLA-based scheduling in distributed systems.
- **Ethical Hacking (NPTEL):** Conducted network security audits and penetration testing within **Kali Linux** environments.
- **Machine Learning (Coursera):** Applied Scikit-Learn for Supervised/Unsupervised Learning and Model Tuning.
- **Industry Simulations (Forage):**
 - Completed AWS Solutions Architecture by designing scalable and resilient cloud environments for key client services.
 - Completed Deloitte Data Analytics by using Python to analyze and visualize large datasets for business intelligence.